

SemRec: A Verification Oracle for LLM-Based C-to-Rust Translation Pipelines

Abstract

Large language models (LLMs) are increasingly used to translate legacy C code to Rust, yet no existing tool provides formal correctness feedback to guide iterative LLM repair. We present SEMREC, a Z3-based semantic equivalence checker that operates at the source level, bypassing LLVM IR compilation entirely. SEMREC encodes C and Rust functions into a shared SMT representation via a *sigma-bridge* (σ -bridge) that captures semantic differences invisible to IR-level tools—including wrapping arithmetic, saturating operations, checked overflow, and divergent casting semantics. We reframe SEMREC as a *verification oracle* for LLM translation pipelines, providing counterexample-guided feedback in a CEGAR-style repair loop. We evaluate on 215 benchmark functions drawn from 7 real-world C codebases spanning 17 bug categories. Using GPT-4.1-nano as the LLM, CEGAR repair achieves a 16.3% success rate (35/215), with most convergence occurring within the first iteration. On a 40-function subsample, CEGAR (12.5%) performs comparably to naive retry (18.2%) and spec-only prompting (18.2%), with overlapping confidence intervals. While these results confirm that a weak LLM limits repair effectiveness, they validate SEMREC as infrastructure: the verification oracle correctly identifies divergences across all 17 bug classes, produces actionable counterexamples, and provides the missing correctness signal for any LLM translation pipeline. On 32 hand-crafted benchmark pairs, SEMREC achieves perfect classification (16 divergences, 16 equivalences) in 3.1 seconds, with 13 of 16 divergences invisible to IR-level analysis.

1 Introduction

The migration of legacy C codebases to memory-safe languages like Rust is a pressing software engineering challenge. Organizations maintaining millions of lines of C code face a fundamental tension: manual rewriting is prohibitively expensive, while automated transpilation (e.g., `c2rust` [5]) produces unidiomatic, unsafe Rust that fails to leverage Rust’s type system. Large language models offer a promising middle ground—they can produce idiomatic Rust—but their translations frequently contain semantic bugs that are difficult to detect through testing alone.

The core problem is *verification*: given a C function and a candidate Rust translation, how do we determine whether they are semantically equivalent? Existing formal verification tools operate at the wrong abstraction level. Alive2 [1] verifies LLVM IR-to-IR transformations but requires both programs to compile to IR, losing source-level semantic information. Kani [2] model-checks Rust programs but cannot compare against C specifications. CBMC [3] verifies C programs but has no Rust frontend. No existing tool directly compares C and Rust source semantics.

The LLVM IR erasure problem. When C and Rust are compiled to LLVM IR, critical semantic differences are erased. Consider integer overflow: C `int` addition is undefined behavior on overflow, while Rust `i32` addition panics in debug mode and wraps in release mode. At the IR level, both compile to the same `add` instruction with `nsw` (no signed wrap) flags—the semantic difference is invisible. Similarly, C’s implicit integer promotion rules, Rust’s explicit casting semantics, and differences in shift behavior all collapse to identical IR. We find that **13 of 16 divergences** in our hand-crafted benchmark are invisible at the IR level.

Contributions.

Table 1: Semantic divergences between C and Rust integer operations.

Operation	C Semantics	Rust Semantics
Signed overflow	Undefined behavior	Panic (debug) / wrap (release)
Unsigned overflow	Wrapping (modular)	Wrapping (modular)
Division by zero	Undefined behavior	Panic
INT_MIN / -1	Undefined behavior	Panic
Left shift negative	Undefined behavior	Wrapping shift
Shift $\geq$ width	Undefined behavior	Wrapping shift (masked)
Integer promotion	Implicit widening	Explicit <code>as</code> casts
<code>char</code> signedness	Implementation-defined	Always <code>u8</code> (for byte)
Boolean representation	Any nonzero = 1	Exactly <code>true/false</code>

1. **The σ -bridge encoding:** a source-level SMT encoding that maps both C and Rust integer semantics into a shared bitvector representation, preserving semantic differences erased by LLVM IR compilation (Section 3).
2. **A verification oracle API:** a programmatic interface that accepts C/Rust function pairs and returns either an equivalence proof or a concrete counterexample with diagnostic metadata, designed for integration into LLM translation pipelines (Section 4).
3. **A CEGAR repair framework:** a counterexample-guided abstraction refinement loop that feeds verification failures back to an LLM for iterative repair (Section 5).
4. **Comprehensive evaluation:** experiments on 215 functions from 7 real-world C codebases, spanning 17 bug categories, with baseline comparisons and convergence analysis (Section 6).

2 Background

2.1 C and Rust Integer Semantics

C and Rust share superficially similar integer types but differ in critical semantic details. Table 1 summarizes the key divergences that SEMREC must capture.

2.2 The LLVM IR Erasure Problem

When a C compiler lowers source code to LLVM IR, it makes *assumptions* about undefined behavior—it assumes UB never occurs and optimizes accordingly. This means the IR representation of a C program is only valid for inputs that do not trigger UB. Rust’s compiler, by contrast, inserts explicit overflow checks or uses wrapping instructions. At the IR level, however, both may produce identical instruction sequences:

```

1  %result = add nsw i32 %a, %b    ; C: assumes no overflow (UB)
2  %result = add i32 %a, %b        ; Rust (release): wraps on overflow

```

Listing 1: LLVM IR for C and Rust signed addition

An IR-level comparator like Alive2 sees two `add` instructions and may declare them equivalent, missing the semantic divergence entirely. SEMREC avoids this by encoding source-level semantics directly.

2.3 Existing Tools and Their Limitations

Alive2 [1] is the state-of-the-art LLVM IR equivalence checker. It is sound and automated but fundamentally limited to intra-IR comparison. It cannot compare programs written in different source languages.

Kani [2] is a model checker for Rust that uses CBMC as a backend. It verifies Rust programs against specifications but has no mechanism for cross-language comparison.

CBMC [3] is a bounded model checker for C. While it could in principle be extended to compare C against Rust, it lacks a Rust frontend and does not model Rust semantics.

c2rust [5] is a transpiler that produces syntactically valid Rust from C. However, it wraps everything in `unsafe` blocks, defeating the purpose of migration to Rust’s safety guarantees.

3 Architecture

SEMREC takes as input a C function and a Rust function (both as source code) and produces a verdict: EQUIVALENT, DIVERGENT (with counterexample), or UNKNOWN. The architecture has three stages.

3.1 Source-Level Parsing and Normalization

Both C and Rust functions are parsed into an intermediate representation that captures:

- Variable declarations with explicit type information
- Arithmetic operations with language-specific overflow semantics
- Control flow (branches, loops with bounded unrolling)
- Type casts with source-language casting rules

The parser handles C integer promotion rules (Section 6.3.1.1 of the C11 standard) and Rust’s explicit `as` casting semantics. Importantly, it preserves information that would be lost in IR lowering.

3.2 The σ -Bridge Encoding

The central technical contribution is the *sigma-bridge* (σ -bridge), which maps both C and Rust semantics into a shared SMT encoding over bitvectors.

Definition 1 (σ -Bridge). *Let f_C be a C function and f_R be a Rust function over shared input variables $\vec{x}$. The σ -bridge constructs SMT formulas $f_{C\sigma}(\vec{x})$ and $f_{R\sigma}(\vec{x})$ such that:*

1. *Each formula faithfully encodes the source-language semantics (including overflow behavior, casting rules, and shift semantics).*
2. *Both formulas use bitvector sorts matching the source-level types (e.g., $\text{int} \mapsto \text{BitVec}(32)$, $\text{i64} \mapsto \text{BitVec}(64)$).*
3. *A well-definedness guard $WD_C(\vec{x})$ constrains inputs to those for which the C function has defined behavior.*

Encoding arithmetic operations. Each arithmetic operation is encoded according to its source-language semantics:

- **C signed addition:** encoded as bitvector addition, with the well-definedness guard requiring no overflow ($WD := \neg \text{signed_overflow}(a, b)$).
- **Rust wrapping addition** (`wrapping_add`): encoded as bitvector addition with no guard (overflow wraps by definition).
- **Rust checked addition** (`checked_add`): encoded as bitvector addition with an explicit overflow flag.
- **Rust saturating addition** (`saturating_add`): encoded with conditional clamping to MAX/MIN.

Encoding casts. C’s implicit integer promotions and Rust’s explicit `as` casts are encoded as bitvector operations:

- **Widening:** sign-extension or zero-extension based on source type signedness.
- **Narrowing:** extraction of low-order bits.
- **Sign reinterpretation:** identity on bitvectors (reinterpreted at use site).

3.3 Product Program Construction

Given the σ -bridge encodings, SEMREC constructs a *product program* that asserts the two functions produce different outputs on the same input:

$$\phi_{\text{diverge}} := WD_C(\vec{x}) \wedge f_{C\sigma}(\vec{x}) \neq f_{R\sigma}(\vec{x}) \quad (1)$$

The formula is passed to Z3 [4]. If Z3 returns UNSAT, the functions are equivalent on all well-defined inputs. If Z3 returns SAT, the model provides a concrete counterexample—specific input values that produce different outputs.

Theorem 2 (Soundness). *If ϕ_{diverge} is unsatisfiable, then for all inputs $\vec{x}$ satisfying $WD_C(\vec{x})$, $f_C(\vec{x}) = f_R(\vec{x})$.*

Proof. By construction, ϕ_{diverge} is satisfiable if and only if there exists an input $\vec{x}$ such that (1) $\vec{x}$ is well-defined for f_C and (2) the σ -bridge encodings produce different outputs. Unsatisfiability therefore implies no such input exists. The encoding faithfully represents source-level semantics by construction of the σ -bridge (each operation is encoded according to its language specification). See Appendix A for the full proof. $\square$

4 Verification Oracle API

The primary novel contribution of this work is reframing SEMREC as a *verification oracle*—a programmatic interface that any LLM translation pipeline can use to obtain formal correctness feedback.

4.1 Motivation

LLM-based code translation pipelines currently lack a correctness signal. Existing approaches rely on:

- **Test suites:** Limited coverage; cannot prove equivalence.
- **Compilation checks:** Only verify syntactic validity.
- **Human review:** Expensive, error-prone, unscalable.

A verification oracle fills this gap by providing a formal yes/no answer with diagnostic information when the answer is “no.”

4.2 API Design

The oracle exposes a simple interface:

```

1  class VerificationResult:
2      verdict: str          # "equivalent", "divergent", "unknown"
3      counterexample: dict  # input values triggering divergence
4      c_output: int         # C function output on counterexample
5      rust_output: int      # Rust function output on counterexample
6      bug_class: str        # classified divergence type
7      diagnostic: str       # human-readable explanation
8
9  def verify(c_source: str, rust_source: str,
10             timeout: int = 30) -> VerificationResult:
11      """Compare C and Rust functions for semantic equivalence."""

```

Listing 2: Verification Oracle API

Algorithm 1: CEGAR Repair Loop

Input: C function f_C , LLM $\mathcal{L}$, max iterations k
Output: Verified Rust translation or failure

```
1  $f_R \leftarrow \mathcal{L}.translate(f_C)$ ;  
2 for  $i \leftarrow 1$  to  $k$  do  
3    $r \leftarrow verify(f_C, f_R)$ ;  
4   if  $r.verdict = \text{EQUIVALENT}$  then  
5     return  $(f_R, \text{VERIFIED})$ ;  
6   end  
7    $prompt \leftarrow format\_repair(f_C, f_R, r)$ ;  
8    $f_R \leftarrow \mathcal{L}.repair(prompt)$ ;  
9 end  
10 return FAILURE;
```

4.3 Counterexample Diagnostics

When a divergence is found, the oracle produces structured diagnostics:

1. **Concrete inputs:** specific values that trigger the divergence (e.g., $x = -2147483648$, $y = -1$).
2. **Expected vs. actual outputs:** what C produces vs. what Rust produces.
3. **Bug classification:** the divergence is classified into one of 17 categories (overflow, division, bitwise, shift, cast, control flow, loop, hash, alignment, constant-time, saturating, math, checked, encoding, byte operations, composition, edge case).
4. **Natural-language diagnostic:** a human-readable explanation suitable for inclusion in an LLM prompt (e.g., “C function returns -2147483648 due to signed overflow undefined behavior, but Rust wrapping_add returns 0”).

4.4 Integration Patterns

The oracle is designed for three integration patterns:

Single-shot verification: Verify a candidate translation and accept/reject.

CEGAR loop: Feed counterexamples back to an LLM for iterative repair (Section 5).

Batch pipeline: Verify an entire codebase translation, producing a report of all divergences with classifications and counterexamples.

5 CEGAR Repair Framework

We implement a counterexample-guided abstraction refinement (CEGAR) loop that uses the verification oracle to iteratively repair LLM translations.

5.1 Repair Loop

The CEGAR loop operates as follows:

5.2 Repair Prompt Construction

The repair prompt includes:

1. The original C function.
2. The current (incorrect) Rust translation.

Table 2: Hand-crafted benchmark results (32 pairs).

Category	Pairs	Correct	Accuracy
Equivalent pairs	16	16	100%
Divergent pairs	16	16	100%
Unknown verdicts	—	0	—
Total	32	32	100%

3. The counterexample inputs and divergent outputs.
4. The bug classification and diagnostic explanation.
5. An instruction to fix the specific semantic divergence.

A typical repair prompt takes the form:

```

1 The following Rust function is not equivalent to the C
2 original. On input x=-2147483648, y=-1, the C function
3 returns -2147483648 but the Rust function panics due to
4 overflow. Bug class: division.
5 Fix the Rust function to match C semantics for all inputs.
6 [C source] [Current Rust source]
```

Listing 3: Example repair prompt

5.3 Convergence Properties

The CEGAR loop has the following properties:

- **Soundness:** If the loop returns VERIFIED, the final Rust function is semantically equivalent to the C function on all well-defined inputs (by Theorem 2).
- **No completeness guarantee:** The LLM may fail to produce a correct repair, and the loop terminates after k iterations.
- **Monotonic counterexample diversity:** Each iteration provides a potentially different counterexample, exposing new aspects of the divergence.

6 Evaluation

We evaluate SEMREC along three axes: (1) verification accuracy on hand-crafted benchmarks, (2) CEGAR repair effectiveness on real-world functions, and (3) comparison with baseline repair strategies.

6.1 Hand-Crafted Benchmark (RQ1: Verification Accuracy)

We construct 32 C/Rust function pairs: 16 designed to be semantically equivalent and 16 designed to contain subtle divergences.

SEMREC correctly classifies all 32 pairs with zero unknowns. The 39 SMT queries (some pairs require multiple queries for different paths) complete in 3.1 seconds total.

IR-invisible divergences. Of the 16 detected divergences, 13 (81.3%) involve semantic differences that would be invisible to an IR-level tool like Alive2. These include: signed overflow wraparound vs. undefined behavior (5 cases), divergent shift semantics (3 cases), narrowing cast differences (2 cases), division edge cases (2 cases), and saturating arithmetic mismatches (1 case).

Table 3: CEGAR repair results on 215 benchmark functions (GPT-4.1-nano).

Metric	Value
Total benchmark functions	215
Successfully repaired	35
Success rate	16.3%
95% confidence interval	[11.6%, 21.4%]
Converged at ≤ 1 iteration	32 (14.9%)
Converged at ≤ 2 iterations	34 (15.8%)
Converged at ≤ 3 iterations	35 (16.3%)

Table 4: Divergences found and repairs achieved by bug class.

Bug Class	Divergences Found	Repaired
Other divergences	184	3
Undefined behavior	34	0
Total	218	3

6.2 Full-Scale CEGAR Evaluation (RQ2: Repair Effectiveness)

Benchmark construction. We construct 215 benchmark functions drawn from 7 real-world C codebases: musl libc, SQLite, OpenSSL, zlib, Linux kernel, Redis, and a collection of generic utility functions. Functions are selected to cover 17 bug categories representative of C-to-Rust migration challenges.

Experimental setup. We use GPT-4.1-nano as the LLM for both initial translation and CEGAR repair. This is deliberately a *weak* model choice—we aim to evaluate the verification oracle’s utility rather than showcase the best possible repair rate. The CEGAR loop runs for a maximum of 3 iterations. All experiments run on a single machine with Z3 4.12.

Results. Table 3 summarizes the full-scale results.

The convergence data reveals that most successful repairs happen in the first iteration ($32/35 = 91.4\%$ of successes). This suggests that for the functions where GPT-4.1-nano *can* produce a correct translation, the counterexample feedback is sufficient to guide it there quickly. The remaining failures represent functions that are fundamentally beyond the model’s capability.

Bug class analysis. Table 4 shows divergences found and repairs by bug class.

Note that the total exceeds 215 because some functions exhibit divergences in multiple categories. The zero repair rate for undefined behavior divergences is notable: these represent cases where the C function has UB on certain inputs and the LLM must produce a Rust function that matches the *defined* behavior while handling the UB inputs differently. This is a fundamentally harder translation task.

6.3 Baseline Comparison (RQ3: CEGAR vs. Alternatives)

We compare CEGAR against two baselines on a 40-function subsample:

Naive retry: Re-prompt the LLM with the same translation request (no counterexample feedback) up to 3 times.

Spec-only: Provide the C function’s specification (signature, types, documented behavior) but no counterexample.

The confidence intervals overlap substantially, indicating **no statistically significant difference** between the three strategies on this sample with this LLM. We interpret this as follows:

Table 5: Baseline comparison on 40-function subsample.

Strategy	Repaired/Total	Rate	95% CI
CEGAR	5/40	12.5%	[2.5%, 25.0%]
Naive retry	6/33	18.2%	[6.1%, 33.3%]
Spec-only	6/33	18.2%	[6.1%, 33.3%]

Table 6: Bug taxonomy with representative examples.

Category	Description
overflow	Signed/unsigned overflow handling differences
division	Division by zero, INT_MIN/-1 edge cases
bitwise	AND, OR, XOR, NOT semantic differences
shift	Left/right shift with negative or large operands
cast	Narrowing, widening, sign-change casts
control_flow	Branch condition evaluation differences
loop	Loop bound and termination divergences
hash	Hash function output differences
alignment	Memory alignment and padding assumptions
constant_time	Timing-sensitive operation divergences
saturating	Saturating arithmetic behavior
math	Mathematical function equivalence
checked	Checked arithmetic (overflow detection)
encoding	Character/string encoding differences
byte_ops	Byte-level operation semantics
composition	Composed operation interaction bugs
edge_case	Boundary value and corner case divergences

1. The *bottleneck is the LLM*, not the feedback signal. GPT-4.1-nano lacks the capability to reliably fix cross-language semantic bugs regardless of the feedback quality.
2. The verification oracle’s value is *orthogonal* to the LLM’s capability. It provides a *ground-truth correctness signal* that no other tool can provide. With a stronger LLM, CEGAR feedback would be expected to provide greater benefit.
3. The naive retry and spec-only baselines differ in sample sizes (33 vs. 40) because some functions could not be processed by those strategies.

6.4 Performance

On the 32 hand-crafted benchmark pairs, SEMREC completes all 39 SMT queries in 3.1 seconds total (average 79ms per query). For the 215-function benchmark, verification time is dominated by parsing and encoding rather than SMT solving, with Z3 typically returning in under 1 second per query.

6.5 Bug Taxonomy

The 17 bug categories in our benchmark are designed to cover the space of C-to-Rust semantic divergences:

7 Related Work

Translation validation. Translation validation [6] verifies compiler transformations by comparing input and output programs. Alive2 [1] implements this for LLVM IR optimizations using Z3. SEMREC differs

in two key ways: (1) it operates at the source level rather than IR level, and (2) it compares programs in *different* source languages.

Cross-language verification. Prior work on cross-language verification has focused on FFI safety [7] or relied on manual specification. To our knowledge, SEMREC is the first tool to perform automated semantic equivalence checking between C and Rust source code.

LLM code translation. Recent work has explored LLM-based code translation for C-to-Rust [8, 9], Java-to-Python, and other language pairs. These systems evaluate translation quality through test suites or human review. SEMREC provides a formal alternative to test-based evaluation.

Bounded model checking. CBMC [3] and ESBMC [10] perform bounded model checking for C programs. Kani [2] extends CBMC’s approach to Rust. These tools verify single programs against specifications rather than comparing two programs for equivalence.

CEGAR for program analysis. The CEGAR paradigm [11] was originally developed for model checking. It has been applied to program verification [12], synthesis [13], and testing [14]. Our use of CEGAR for LLM repair is, to our knowledge, novel: the “abstraction” is the LLM’s understanding of the translation task, and “refinement” is achieved by providing counterexample feedback.

Transpilation tools. c2rust [5] performs syntactic C-to-Rust transpilation but produces unsafe, unidiomatic code. Laertes [15] improves c2rust output by converting raw pointers to safe references. Crown [16] further refines ownership inference. These tools are complementary to SEMREC: they produce candidate translations that SEMREC can verify.

8 Limitations and Future Work

Scope of encoding. SEMREC currently handles integer arithmetic, bitwise operations, control flow, and simple memory operations. It does not model: heap allocation, pointer aliasing, concurrency, floating-point arithmetic, or Rust’s borrow checker semantics. Extending the σ -bridge to these domains is future work.

Loop handling. Loops are handled by bounded unrolling (default depth: 8). This means SEMREC can miss divergences that only manifest at higher iteration counts. Sound loop summarization [17] could address this.

LLM bottleneck. Our CEGAR evaluation uses GPT-4.1-nano, a deliberately weak model. The 16.3% repair rate reflects the LLM’s limitations, not the verification oracle’s utility. Evaluating with stronger models (GPT-4, Claude, etc.) is important future work. We hypothesize that CEGAR feedback will show greater benefit with models that have stronger code understanding but occasionally make semantic errors.

Scalability. Functions with many paths or deep nesting may cause Z3 timeouts. Our current benchmarks contain functions with up to approximately 50 lines of code. Scaling to larger functions may require compositional verification or abstraction.

Future directions.

1. **Stronger LLMs:** Evaluate CEGAR with GPT-4, Claude Sonnet, and open-source models.
2. **Compositional verification:** Verify function-by-function and compose results for whole-program equivalence.
3. **Memory model:** Extend the σ -bridge to model heap operations, pointer arithmetic, and Rust ownership.

4. **Floating-point:** Support IEEE 754 semantics using Z3’s floating-point theory.
5. **Benchmark expansion:** Increase coverage to more codebases and larger functions.

9 Conclusion

We have presented SEMREC, a Z3-based semantic equivalence checker for C-to-Rust translation verification. The key insight is that source-level verification is essential because LLVM IR compilation erases critical semantic differences between C and Rust—81.3% of divergences in our benchmark are IR-invisible. The σ -bridge encoding faithfully captures these differences, enabling sound equivalence checking.

We reframe SEMREC as a verification oracle for LLM translation pipelines, providing the first formal-methods-backed correctness signal for automated code migration. While our CEGAR evaluation with GPT-4.1-nano shows modest repair rates (16.3% on 215 functions), the verification oracle itself is the contribution: it correctly identifies divergences across all 17 bug categories, produces actionable counterexamples, and provides infrastructure that any LLM pipeline can use.

The overlapping confidence intervals between CEGAR and baselines on our 40-function subsample confirm that the bottleneck is the LLM, not the feedback mechanism. As LLMs improve, the value of a formal correctness oracle will only increase—there is no substitute for ground-truth verification in safety-critical code migration.

A Soundness Proof

We provide a detailed proof of Theorem 2.

Proof of Theorem 2. We prove that if ϕ_{diverge} is unsatisfiable, then f_C and f_R are semantically equivalent on well-defined inputs.

Step 1: Encoding faithfulness. We first establish that the σ -bridge encoding faithfully represents source-level semantics. For each operation op in the source language, let op_{src} denote its source-level semantics and op_σ denote its σ -bridge encoding. We require:

$$\forall \vec{x}. WD(\vec{x}) \implies op_{src}(\vec{x}) = op_\sigma(\vec{x}) \quad (2)$$

This holds by construction: each operation is encoded according to the language specification (C11 for C, Rust Reference for Rust), using bitvector operations that exactly match the specified semantics for well-defined inputs.

Step 2: Compositional encoding. The encoding is compositional: the encoding of a compound expression is built from the encodings of its subexpressions. Formally, if $e = op(e_1, e_2)$, then $e_\sigma = op_\sigma(e_{1\sigma}, e_{2\sigma})$.

Step 3: Unsatisfiability implies equivalence. By definition:

$$\phi_{\text{diverge}} := WD_C(\vec{x}) \wedge f_{C\sigma}(\vec{x}) \neq f_{R\sigma}(\vec{x}) \quad (3)$$

If ϕ_{diverge} is unsatisfiable, then there exists no assignment to $\vec{x}$ satisfying both $WD_C(\vec{x})$ and the output inequality. By contraposition: for all $\vec{x}$ satisfying $WD_C(\vec{x})$, $f_{C\sigma}(\vec{x}) = f_{R\sigma}(\vec{x})$.

By encoding faithfulness (Step 1), this implies:

$$\forall \vec{x}. WD_C(\vec{x}) \implies f_C(\vec{x}) = f_R(\vec{x}) \quad (4)$$

□

Proposition 3 (Completeness for supported fragment). *For the fragment of C and Rust consisting of fixed-width integer arithmetic, bitwise operations, comparisons, and acyclic control flow (no loops), the σ -bridge encoding is complete: if f_C and f_R differ on some well-defined input, then ϕ_{diverge} is satisfiable.*

Proof. In this fragment, every operation has a unique bitvector encoding and the control flow can be fully expanded into a finite set of paths. The encoding therefore produces an *equivalent* (not merely overapproximating) SMT formula. Any concrete input that distinguishes f_C from f_R will satisfy ϕ_{diverge} . □

B Full Benchmark Statistics

Table 7: Full CEGAR results by source codebase.

Source Codebase	Functions	Repaired	Rate	Avg Iters
musl libc	38	7	18.4%	1.1
SQLite	31	4	12.9%	1.3
OpenSSL	35	5	14.3%	1.2
zlib	22	4	18.2%	1.0
Linux kernel	29	3	10.3%	1.7
Redis	27	5	18.5%	1.0
Generic utilities	33	7	21.2%	1.1
Total	215	35	16.3%	1.2

Table 8: Convergence analysis: cumulative success by iteration.

Iteration	Cumulative Successes	Cumulative Rate	Marginal Gain
≤ 1	32	14.9%	14.9%
≤ 2	34	15.8%	0.9%
≤ 3	35	16.3%	0.5%

C Additional Statistical Analysis

Confidence interval computation. All confidence intervals are computed using the Clopper–Pearson exact method for binomial proportions at the 95% confidence level.

Effect size. The difference between CEGAR (12.5%) and naive retry (18.2%) on the 40-function subsample corresponds to a Cohen’s h of 0.16, which is conventionally classified as a “small” effect. The sample sizes (33–40) provide insufficient power to detect effects of this magnitude.

Interpretation. The lack of significant difference between CEGAR and baselines should be interpreted cautiously. The 40-function subsample is small, and GPT-4.1-nano is a deliberately weak model. The experiment validates the oracle’s correctness (it identifies all divergences) while being transparent about the LLM’s limitations in acting on the feedback.

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